

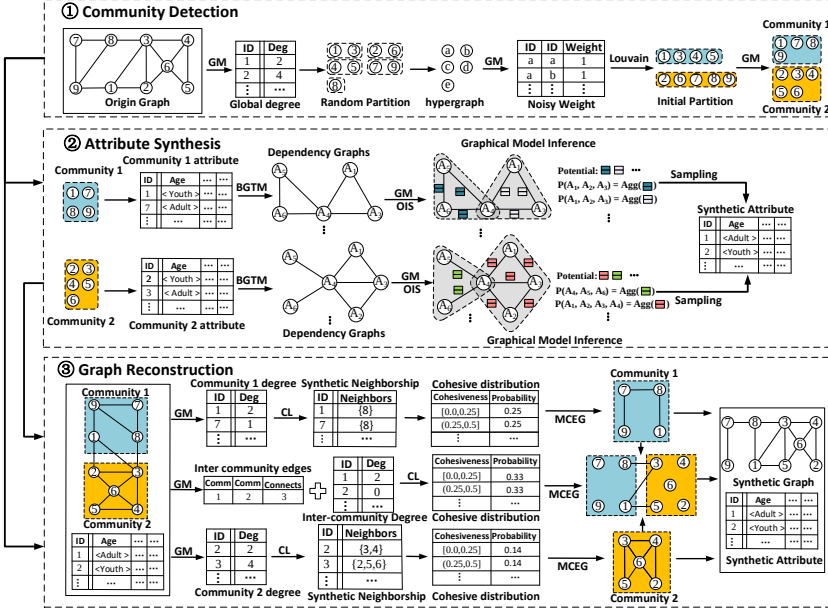


Fig. 2. Framework of PrivAGS

Stage 1: Community Detection. We design a community detection algorithm to partition nodes efficiently. First, we determine the number of supernodes based on graph density and global noisy degree mappings. Nodes are then randomly assigned to supernodes, and noisy connections between supernode pairs form weighted superedges and a weighted graph. Using this weighted graph, PrivAGS performs community partitioning with the Louvain algorithm [4]. Finally, PrivAGS refines the partition using noisy edge connections between nodes and communities. Due to the limited space, further details are provided in [1] Appendix A.

Stage 2: Attribute Synthesis. We use the community partitions obtained from Stage 1 as the basis for attribute synthesis. For each community, we extract the corresponding attribute sets and construct an attribute dependency graph using the Bounded Gaussian Threshold Mechanism (BGTm). After that, we employ the optimized inference structure (OIS) with minimal noise power as the inference structure of the probabilistic graphical model to infer attribute distribution and release node attributes.

Stage 3: Graph Reconstruction. In graph reconstruction, community partitions and synthetic attributes are used with the Gaussian mechanism to extract inter-community node degree mappings within each community. Synthetic neighbor relationships are generated through these mappings and attributes from Stage 2 to calculate the community's cohesiveness distribution. This distribution, combined with the degree mappings, allows MCEG to reconstruct the internal community structure. For the intra-community structure, PrivAGS computes intra-community node degree mappings using global degree mappings from Stage 1 and inter-community mappings. It then generates the required MCEG parameters and applies them to create edges between communities.

Privacy Budget Analysis. Our approach includes three steps: community partitioning, attribute synthesis, and graph reconstruction. The community partitioning uses ϵ_1 . Attribute synthesis involves dependency graph construction (ϵ_2) and attribute generation (ϵ_3), while optimal inference structure requires no budget. Graph reconstruction consumes ϵ_4 . Thus, the total privacy budget is $\epsilon = \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4$.

THEOREM 1. *Our solution satisfies (α, ϵ) -DAE RDP, where $\epsilon = \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4$.*